E-Commerce Store Performance Analysis Report

Monthly Sales and Product Performance Analysis

---

١. Introduction

This report provides a comprehensive analysis of the performance of an e-commerce store based on available data covering monthly sales, product prices, product categories, units sold, product ratings, and review counts.

The analysis focuses on key business performance indicators, including:

- Total revenue
- Monthly revenue performance
- Top-selling products
- Revenue performance by product category
- Monthly sales trends
- Product ratings
- Key business insights and actionable recommendations

The available dataset includes:

- Monthly sales data
- Product names
- Product categories
- Product prices
- Product ratings
- Number of product reviews
- Units sold

Data Limitation

The current dataset does not include detailed order-level or customer-level information such as:

- order_id

- customer_id
- purchase_date

Therefore, metrics such as **Total Orders, Average Order Value (AOV), Customer Retention Rate, Repeat Purchase Rate, and Customer Lifetime Value (CLV)** cannot be calculated accurately from the current dataset.

These metrics can be incorporated in future analyses if detailed transaction and customer data become available.

---

٢. Executive Summary

The analysis shows that the e-commerce store generated total revenue of **1,488,704,183.02** during the analyzed period.

Key findings include:

- **Highest-revenue month:** Month 10, generating **129,100,593.26**
- **Top-selling product:** Product_224, with **9,151 units sold**
- **Highest-revenue category:** Books, generating **236,782,805.60**
- **Total units sold: 6,019,912 units**
- **Top five products combined: 44,544 units sold**
- **Average product price: 247.68**
- **Average product rating: 3.03**

Overall, the store demonstrates strong and relatively stable sales performance. However, the analysis also reveals opportunities to improve inventory management, strengthen marketing strategies, optimize pricing, and improve customer and product experience.

---

٣. Financial Performance Analysis

٣٫١ Total Revenue

The total recorded revenue during the analyzed period was:

١,٤٨٨,٧٠٤,١٨٣٫٠٢

Revenue was calculated by multiplying the number of units sold by the corresponding product price and aggregating the results across the available data.

This figure provides a broad overview of the store's overall commercial activity and represents the primary financial performance indicator available in the dataset.

٣٫٢ Monthly Revenue Performance

Monthly revenue was calculated by multiplying the units sold by the corresponding product prices and aggregating the results for each month.

The monthly revenue performance was as follows:

Month	Revenue
Month 10	١٢٩,١٠٠,٥٩٣٫٢٦
Month 4	١٢٦,٦٣١,٤٠٧٫٣١
Month 11	١٢٥,٦١٧,١٩٤٫٩٨
Month 2	١٢٥,٦١٣,٠٣٣٫٩١
Month 12	١٢٤,٧٠١,٨٩٢٫٠٦
Month 3	١٢٤,٥٢٢,١٢٦٫٣٦
Month 7	١٢٤,٣٩٧,٩٤٨٫٩٨
Month 8	١٢٣,٠٨٢,٥٥٧٫٧٣
Month 1	١٢٢,٠٠٠,٧٢٠٫٥٤
Month 5	١٢١,٩١٥,٣١٨٫١٣
Month 9	١٢١,٤٩٩,٢٠٦٫١٦
Month 6	١١٩,٦٢٢,١٩٧٫٨٨

Key Finding

Monthly revenue remained relatively stable throughout the analyzed period, with **Month 10 recording the highest revenue**.

This indicates that the store has a relatively consistent revenue base, while the peak performance in Month 10 represents an opportunity for further investigation.

The company should examine whether the Month 10 increase was influenced by:

- Marketing campaigns
- Promotional offers
- Seasonal demand
- Pricing changes
- Product availability
- External market factors

Understanding these drivers could help the business reproduce similar performance during other periods.

---

٤. Product Performance Analysis

٤٫١ Top-Selling Products

The following products recorded the highest total number of units sold:

Rank	Product	Units Sold
١	Product_224	٩,١٥١
٢	Product_286	٨,٩٢١
٣	Product_734	٨,٩١٤
٤	Product_905	٨,٧٨٣
٥	Product_180	٨,٧٧٥
٦	Product_853	٨,٧٦٥
٧	Product_239	٨,٧٢٤
٨	Product_924	٨,٥٢٥
٩	Product_937	٨,٤٥٩

Rank	Product	Units Sold
١٠	Product_197	٨,٤١٨

Key Finding

The top-selling products represent important strengths within the current product portfolio.

Products such as **Product_224, Product_286, and Product_734** demonstrate particularly strong customer demand. Maintaining adequate inventory levels and strategically promoting these products could help the company protect sales and maximize revenue opportunities.

٤,٢ Combined Performance of the Top Five Products

The top five products generated a combined:

٤٤,٥٤٤ units sold

This indicates that a significant portion of demand is concentrated among a relatively small group of products.

Therefore, the company should consider:

- Maintaining sufficient inventory for high-demand products
- Prioritizing these products in marketing campaigns
- Improving their visibility on the e-commerce platform
- Using cross-selling and bundle offers
- Monitoring their availability closely

---

٥. Category Performance Analysis

Revenue was aggregated across the different product categories as follows:

Rank	Category	Revenue
١	Books	٢٣٦,٧٨٢,٨٠٥,٦٠
٢	Sports	٢٣٢,٦٤٨,٨٢٦,١١

Rank	Category	Revenue
٣	Toys	٢٣٠,٢٣٧,١٦٧,١٩
٤	Health	٢٢١,٧٣٦,٩٢٦,٧٧
٥	Electronics	٢٠١,٦٧٤,٧٤٤,٣٦
٦	Clothing	١٨٧,٢٥٨,٣٤١,٤٤
٧	Home & Kitchen	١٧٨,٣٦٥,٤٨٩,٧٢

Key Finding

Books is the highest-revenue category, generating **236,782,805.60**.

The categories **Sports, Toys, and Health** also demonstrate strong performance and represent attractive opportunities for additional marketing investment and product expansion.

Meanwhile, **Clothing** and **Home & Kitchen** generate comparatively lower revenue and may require additional promotional or pricing strategies to improve their performance.

---

٦. Monthly Sales Trend Analysis

The monthly sales trend was analyzed by comparing the number of units sold in each month with the previous month.

The analysis identified several periods of decline, including:

- **Month 5:** Largest decrease in units sold compared with the previous month
- **Month 9:** Noticeable decline
- **Month 11:** Moderate decline
- **Month 12:** Noticeable decline

Key Finding

The fluctuations in monthly unit sales may indicate the presence of seasonal demand patterns or changes in customer purchasing behavior.

To better understand these fluctuations, the company should investigate:

- Marketing campaign performance

- Promotional activities
- Seasonal purchasing patterns
- Inventory availability
- Pricing strategies
- Product demand by category

Identifying the causes of weaker months could help the business develop more effective seasonal planning and sales strategies.

---

٧. Key Business Insights

The analysis generated several important business insights:

١. Strong Overall Revenue

The store generated total revenue of **1,488,704,183.02**, indicating a strong level of commercial activity during the analyzed period.

٢. Month ١٠ Was the Strongest Revenue Period

Month 10 generated the highest monthly revenue at **129,100,593.26**, making it an important period for further investigation.

٣. Product_٢٢٤ Is the Top-Selling Product

Product_224 recorded the highest number of units sold, with **9,151 units**, making it one of the most important products in the current portfolio.

٤. Books Is the Leading Revenue Category

The **Books** category generated **236,782,805.60**, making it the highest-performing category by revenue.

٥. Demand Is Concentrated Among Leading Products

The top five products generated **44,544 units**, indicating that demand is relatively concentrated among a limited number of products.

٦. High Sales Volume Requires Effective Inventory Management

The store sold a total of **6,019,912 units**, representing a substantial operational volume and highlighting the importance of effective inventory planning.

٧. Average Product Price

The average product price is **247.68**, indicating that the store's product portfolio includes a mix of relatively mid-priced and higher-priced products.

٨. Product Ratings Present an Opportunity for Improvement

The average product rating is **3.03**, which represents a moderate level of customer satisfaction and suggests potential opportunities to improve product quality and the overall customer experience.

٩. Customer-Level Analysis Is Currently Limited

The available data does not contain sufficient order-level or customer-level information to accurately calculate important customer and transaction metrics such as:

- Average Order Value (AOV)
- Customer Retention Rate
- Repeat Purchase Rate
- Customer Lifetime Value (CLV)

١٠. Stable Monthly Revenue

The relatively stable monthly revenue suggests that the store maintains a consistent level of commercial activity, while the weaker months present opportunities for targeted improvement.

---

٨. Strategic Recommendations

٨,١ Increase Investment in the Books Category

Books is currently the highest-revenue category.

The company could consider:

- Increasing marketing investment in this category
- Running targeted advertising campaigns
- Introducing seasonal promotions
- Creating product bundles
- Offering targeted discounts

- Expanding the product selection

The objective should be to capitalize on existing demand while maintaining profitability.

٨,٢ Increase Inventory for Top-Selling Products

Products such as **Product_224, Product_286, and Product_734** demonstrate strong demand.

The company should:

- Maintain sufficient stock levels
- Monitor inventory continuously
- Reduce the risk of stockouts
- Prioritize high-demand products in procurement
- Use demand trends to improve inventory forecasting

Ensuring product availability can help prevent lost sales opportunities.

٨,٣ Analyze Month ١٠ Performance

Since Month 10 generated the highest revenue, the company should conduct a detailed analysis to determine what drove this performance.

Potential factors include:

- Marketing campaigns
- Discounts and promotions
- Seasonal demand
- Product availability
- Pricing changes
- Increased customer traffic

Once the main drivers are identified, similar strategies could potentially be applied during other periods with comparable market conditions.

٨,٤ Expand High-Performing Categories

Categories such as:

- Sports
- Toys
- Health

also demonstrate strong revenue performance.

The company could consider:

- Expanding the product range
- Introducing complementary products
- Increasing marketing investment
- Improving category visibility
- Developing category-specific promotional campaigns

٨,٥ Improve the Performance of Lower-Revenue Categories

Categories such as **Clothing** and **Home & Kitchen** generate comparatively lower revenue.

Potential strategies include:

- Targeted discounts
- Product bundles
- Promotional campaigns
- Cross-selling
- Improved product presentation
- Competitive pricing analysis

These categories may represent additional growth opportunities if supported by appropriate strategies.

٨,٦ Improve Product Quality and Customer Experience

The average product rating of **3.03** indicates room for improvement.

The company should consider reviewing:

- Product quality
- Packaging
- Delivery performance
- Customer service
- Product descriptions
- Return and refund processes

Customer feedback should also be collected and analyzed regularly to identify recurring issues and improve satisfaction.

٨,٧ Develop a More Comprehensive Customer and Order Data System

For more advanced business analysis, the company should develop a structured system for collecting order and customer-level information.

Important fields should include:

- Order ID
- Customer ID
- Purchase Date
- Product ID
- Quantity
- Unit Price
- Discount
- Order Value
- Customer Location
- Payment Method

This would allow the company to calculate advanced KPIs such as:

- Total Orders

- Average Order Value (AOV)
- Customer Retention Rate
- Repeat Purchase Rate
- Customer Lifetime Value (CLV)
- Customer Acquisition Cost (CAC)

٨,٨ Build a Regular Performance Dashboard

A business intelligence dashboard should be developed to provide management with continuous visibility into business performance.

The dashboard should monitor:

- Monthly revenue
- Units sold
- Top-selling products
- Revenue by category
- Monthly growth
- Product ratings
- Inventory performance
- Key business KPIs

Regular monitoring would allow management to identify changes quickly and make more data-driven decisions.

٨,٩ Re-Evaluate Pricing Strategies

A dedicated pricing analysis could be conducted across different categories and products.

The objective should be to identify the optimal balance between:

- Price
- Demand
- Revenue

- Profitability
- Customer value
- Market competitiveness

This analysis could help determine whether certain products are underpriced or overpriced relative to their demand and market position.

٨,١٠ Develop a Long-Term Growth Strategy

The current analysis provides a foundation for a data-driven growth strategy focused on:

- Strengthening high-performing categories
- Increasing inventory for high-demand products
- Improving weaker product categories
- Enhancing customer experience
- Optimizing pricing
- Expanding product offerings
- Improving data collection
- Developing advanced business intelligence capabilities

---

٩. Conclusion

The analysis of the e-commerce store's performance indicates a strong overall level of commercial activity, with total revenue reaching **1,488,704,183.02** and total unit sales reaching **6,019,912 units**.

The analysis identified several important performance drivers. **Books** was the highest-revenue category, **Product_224** was the top-selling product, and **Month 10** recorded the highest monthly revenue.

The results also indicate that demand is concentrated among a limited number of products and categories, creating clear opportunities for more effective inventory management and targeted marketing.

At the same time, the average product rating of **3.03** highlights an opportunity to improve product quality and customer experience. The relatively lower performance of categories

such as **Clothing** and **Home & Kitchen** also provides potential areas for targeted growth initiatives.

However, the current dataset has limitations because it does not contain detailed order and customer information. As a result, important metrics such as **Average Order Value, Customer Retention, Repeat Purchase Rate, and Customer Lifetime Value** cannot currently be calculated accurately.

Moving forward, the company should focus on strengthening its highest-performing products and categories, improving inventory and pricing strategies, enhancing customer experience, and building a more comprehensive data infrastructure.

By adopting a continuous, data-driven approach to decision-making, the company can improve operational efficiency, increase profitability, identify new growth opportunities, and strengthen its competitive position in the e-commerce market.